Valid and efficient possibilistic inference across a spectrum of partial prior information[1]

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[1] Working paper: https://researchers.one/articles/21.05.00001
[2] Research partially supported by the NSF, SES–2051225
[3] www4.stat.ncsu.edu/~rgmarti3

Introduction

- *Bayesian, frequentist, fiducial, ...*
- That there are a variety of approaches is great for us as academics, but not for science/society/etc
- There can't be a one-size-fits-all statistical *solution*
- But can the *approach* be one-size-fits-all?
- We all want "valid uncertainty quantification"
- A broader view of probability makes unification possible[4]

[4] ISIPTA'19: "There's more to uncertainty than probability"

Intro, cont.

- We tend to think in binary terms:
  - prior $\iff$ Bayesian
  - no prior $\iff$ classical/frequentist
- *Key point:* frequentist isn't "no prior" — it's *every prior*
- So, there's a spectrum indexed by prior information

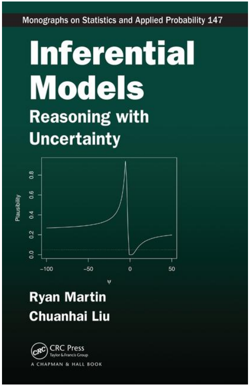
- Understanding this spectrum will lead to, e.g.,
  - meaningful notions of unification
  - new & improved methods

This talk

- a bit of background on *every prior*
- partial priors
- partial prior validity & its consequences
- how to achieve partial prior validity
- quick illustration
- perspectives, e.g., on high-dim problems

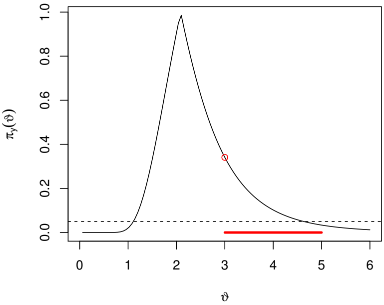
Background: *every prior*

- $Y \sim \mathsf{P}_\theta$, $\theta \in \mathbb{T}$
- *Inferential models* (IMs)
  - $y \mapsto (\underline{\Pi}_y, \overline{\Pi}_y)$, lower and upper prob's on $\mathbb{T}$
  - degrees of belief
  - driven by random sets
  - frequentist guarantees
- valid uncertainty quantification
  - Fisher's *disjunction* is key
  - calibrated "probabilities"



Background, cont.

- IMs & *possibility theory**
- IM output determined by a *plausibility contour* function $\pi_y : \mathbb{T} \to [0,1]$
- i.e., $\overline{\Pi}_y(A) = \sup_{\vartheta \in A} \pi_y(\vartheta)$
- provably valid UQ!
- consequences
  - $\Rightarrow$ tests control size
  - $\Rightarrow$ level sets of π_y are confidence regions



*Liu and M., `https://researchers.one/articles/20.08.00004`

Background, cont.

- This theory covers the frequentist end of the spectrum
- *Every procedure* with exact error rate control can be represented as a valid IM/possibility measure[5]
- e.g., p-values are IM plausibility/upper prob's
- Nancy Reid at BFF4 about me: "IMs are the only answer"

RESEARCHERS.ONE researchers.one/articles/21.01.00002

An imprecise-probabilistic characterization of frequentist statistical inference*

Ryan Martin†

January 25, 2021

[5] I presented this at BFF6.5, early version at BFF4

Partial priors

- e.g., "I'm 90% sure that $\theta \in [3, 7]$"
- $\mathscr{Q}$ = a (credal) set of priors compatible with available info
- Prior upper probability, $\overline{\mathsf{Q}}(A) = \sup_{\mathsf{Q} \in \mathscr{Q}} \mathsf{Q}(A)$
- "Upper joint distribution" for (Y, Θ):

$$\overline{\mathsf{P}}_{\mathscr{Q}}(Y \in B,\, \Theta \in A) = \sup_{\mathsf{Q} \in \mathscr{Q}} \underbrace{\int_A \mathsf{P}_\theta(B)\, \mathsf{Q}(d\theta)}_{\text{joint prob wrt Q}}$$

- Notation:
  - Let $\overline{\Pi}_{y,\mathscr{Q}}$ denote the upper prob of a *partial prior IM*
  - IM's *contour* $\pi_{y,\mathscr{Q}}(\vartheta) = \overline{\Pi}_{y,\mathscr{Q}}(\{\vartheta\})$

Partial prior validity

- Priors always introduce bias
- But a partial prior isn't artificial, *so its bias is good*
- Properties we seek should be relative to $\mathscr{Q}$

Definition.

A partial prior IM is *valid* relative to $\mathscr{Q}$ if

$$\overline{\mathsf{P}}_{\mathscr{Q}}\{\overline{\Pi}_{Y,\mathscr{Q}}(A) \leq \alpha, \, \Theta \in A\} \leq \alpha, \quad \text{all } \alpha, \text{ all } A.$$

It's *strongly valid* relative to $\mathscr{Q}$ if

$$\overline{\mathsf{P}}_{\mathscr{Q}}\{\pi_{Y,\mathscr{Q}}(A) \leq \alpha\} \leq \alpha, \quad \text{all } \alpha.$$

Partial prior validity, cont.

- Remarks:
  - Strong validity is stronger than validity
  - Validity wrt $\mathscr{Q} = \{\text{all priors}\}$ reduces to my usual definition
  - Smaller $\mathscr{Q}$ relaxes the validity, creates opportunities to gain in efficiency without sacrificing validity

Theorem.

1. *Behavioral.* "Validity implies no sure loss"
2. *Statistical.* Strong validity implies, e.g., that the $100(1-\alpha)\%$ plausibility region $C_\alpha(y) = \{\vartheta : \pi_{y,\mathscr{Q}}(\vartheta) > \alpha\}$ satisfies

$$\overline{\mathsf{P}}_{\mathscr{Q}}\{C_\alpha(Y) \not\ni \Theta\} \leq \alpha$$

Achieving partial prior validity

- Some "standard" things are valid, e.g., generalized Bayes
- But they're inefficient...
- *Brand new* proposal here, aimed at validity + efficiency

- Partial prior structure in $\mathscr{Q}$:
  - Assume $\overline{\mathrm{Q}}$ is a possibility measure with contour q
  - That is, $\overline{\mathrm{Q}}(A) = \sup_{\vartheta \in A} q(\vartheta)$
  - So far this is general enough
- Likelihood function $L_y(\cdot)$ from the model

Achieving partial prior validity, cont.

- Combine L_y and q in an *almost-familiar* way:

$$h_y(\vartheta) = \frac{L_y(\vartheta)\, q(\vartheta)}{\sup_{t \in \mathbb{T}} L_y(t)\, q(t)}, \quad \vartheta \in \mathbb{T}$$

- Other h functions can be considered, but this one is nice
- Define the partial prior IM's contour function as

$$\boxed{\pi_{y,\mathscr{Q}}(\vartheta) = \overline{\mathsf{P}}_{\mathscr{Q}}\{h_Y(\Theta) \leq h_y(\vartheta)\}}$$

Theorem.

The partial prior IM determined by the contour function defined in the box is strongly valid relative to $\mathscr{Q}$.

Achieving partial prior validity, cont.

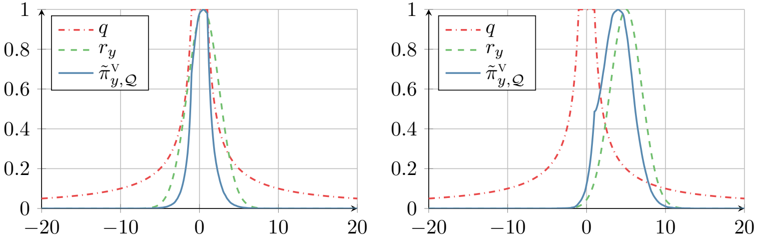
- Quick proof of concept:
  - frequentist/"every prior" case $\iff q \equiv 1$
  - then $h_y(\cdot)$ is the relative likelihood
  - $\pi_{y,\mathscr{Q}}(\cdot) \approx$ p-value function for the likelihood ratio test
- How it works:
  - level sets of h_y determine a *plausibility ordering*
  - q does the usual regularization, helps with efficiency
  - $\overline{\mathsf{P}}_{\mathscr{Q}}$-prob calculation adjusts the size of the level sets
- Computation? It's a Choquet integral, which simplifies to[6]

$$\pi_{y,\mathscr{Q}}(\vartheta) = \int_0^1 \Big[\sup_{\theta : q(\theta) > s} \mathsf{P}_\theta\{h_Y(\theta) \leq h_y(\vartheta)\} \Big] \, ds$$

[6]Since $\overline{\mathsf{Q}}$ is a possibility measure

Illustration

- Simple effect size problem: $Y \sim \mathsf{N}(\theta, \sigma^2)$, σ known
- Partial prior for θ:
  - "I expect $|\theta|$ is no more than 1"
  - i.e., $\mathscr{Q} = \{\mathsf{Q} : \int |\theta| \, \mathsf{Q}(d\theta) \leq 1\}$
  - Markov prior contour: $q(\vartheta) = \min(1, |\vartheta|^{-1})$



Perspectives

- Lots of applications involve partial prior info, we just don't realize/acknowledge it, e.g., ratio-of-means[7]
- Important and common case: *high-dim problems*
- Often there is good scientific reason to believe that high-dim θ has a low-dim structure, e.g., sparsity
- On the methods side, incorporating this structure at either of the two ends of the spectrum is awkward
  - frequentist regularization is ad hoc & anti-frequentist
  - Bayesian regularization requires embellishment
- But it's easy to accommodate as a partial prior

[7] One of Cox's "challenge problems" from BFF4

Perspectives, cont.

- Moreover, there are impossibility theorems that establish an incompatibility between
  - every-prior validity
  - efficiency relative to low-dim structure
- Existing strategy is to get (asymptotic) validity in a parameter space with deceptive parameter values carved out
- Better strategy is to let validity be relative to the assumed sparsity structure, no need for asymptotics or carving
- My theory holds independent of n and $\dim(\theta)$
- Just a question of how to compute the Choquet integral

Perspectives, cont.

Don Fraser

How can a discipline, central to science and to critical thinking, have two methodologies, two logics, two approaches that frequently give substantially different answers to the same problems. Any astute person from outside would say, "Why dont they put their house in order?" ...

- New view — a single approach/logic applied to a problem-specific position on the spectrum
- $(\text{data}, \text{model}, \mathscr{Q}) \longrightarrow \mathscr{Q}$-valid IM
- Pick the smallest $\mathscr{Q}$ you can justify, "skin in the game"

Not just for inference

- https://researchers.one/articles/20.01.00010

Valid inferential models for prediction in supervised learning problems*

Leonardo Cella† and Ryan Martin†

April 5, 2022

- https://researchers.one/articles/21.12.00005

Inferential models and the decision-theoretic implications of the validity property

Ryan Martin*

December 25, 2021

Conclusion

- Key observation: frequentist $=$ *every prior*
- Unification through a spectrum of partial prior info
  - quantify partial prior via $\mathscr{Q}$, $\overline{\mathrm{Q}}$, or q, ...
  - proposed a notion of validity that applies across spectrum
- General construction of a valid partial prior IM
  - $(\text{data}, \text{model}, \mathscr{Q}) \longrightarrow \mathscr{Q}$-valid IM
  - At least sets a baseline for comparison
- Excited about the "possibilities" (pun intended)
  - high-dim problems specifically
  - "putting our house in order" more generally
- Questions:
  - efficient computational strategies??
  - beyond parametric models??
  - ...

The end

Thanks for your attention!

`www4.stat.ncsu.edu/~rgmarti3`

`https://researchers.one/articles/21.05.00001`

Valid and efficient imprecise-probabilistic inference across a spectrum of partial prior information

Ryan Martin*

April 21, 2022